

- 1** A student takes measurements to determine the acceleration of free fall. He throws a ball of mass 0.010 kg vertically upwards with a velocity of 25 m s^{-1} from ground level.

The variation with time t of the vertical velocity v of the ball is shown in Fig. 1.1.

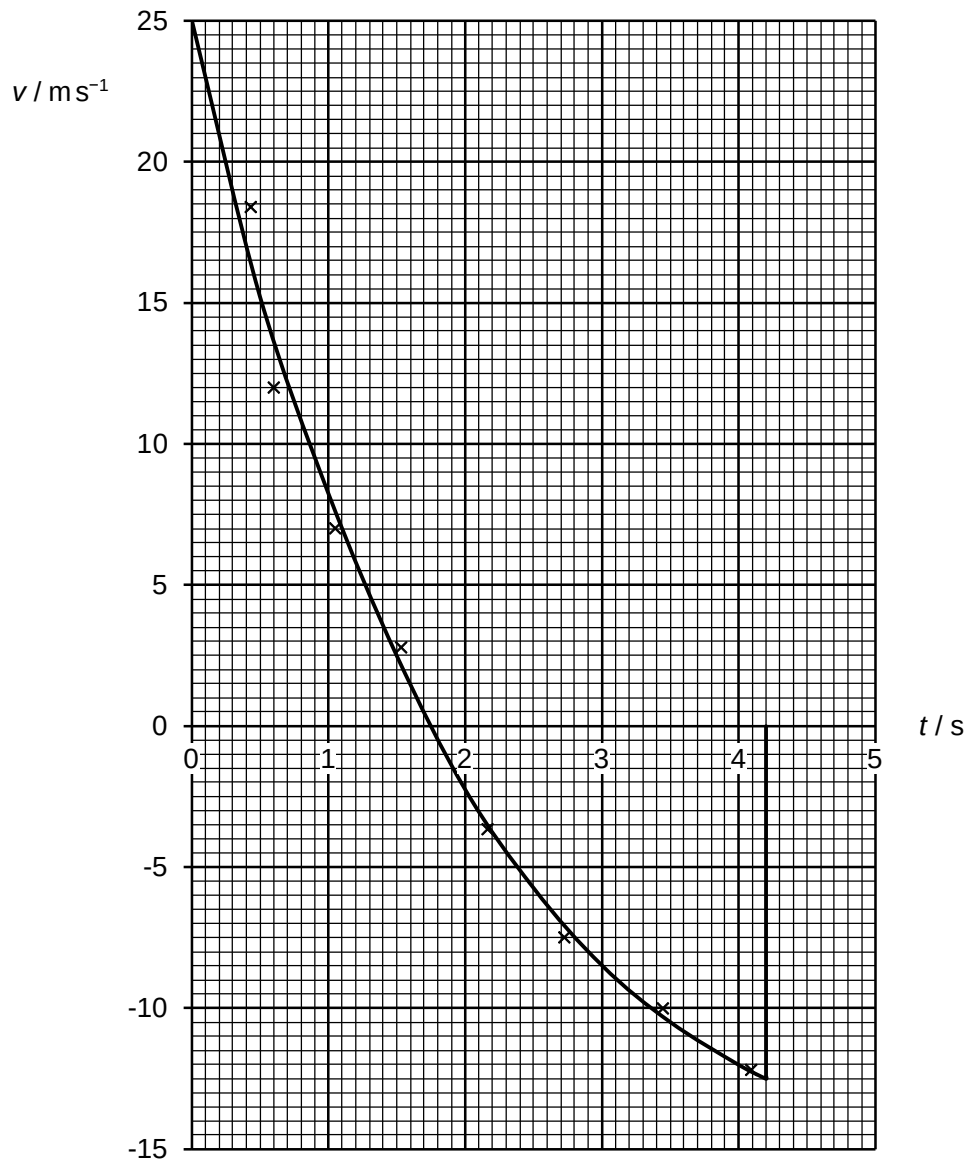


Fig. 1.1

(a) Using Fig. 1.1, explain

(i) how it can be deduced that air resistance is present in this experiment,

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..... [2]

(ii) how the magnitude of the acceleration of free fall can be determined.

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..... [2]

(b) Calculate the magnitude of the maximum force due to air resistance on the ball.

Explain your working.

force = N [3]

2 Fig. 2.1 shows the variation with displacement d of the force F applied to an object at rest